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Q Chapter 1 exam-style questions

أسئلة امتحانية حقيقية: تصنيف مع تبرير، بناء وقراءة، اكتشاف أخطاء. ليست أسئلة مباشرة

Built from the outline's assessment criteria for this chapter. Every question carries its lesson tag and outcome tag; a wrong answer points to the lesson file to reopen. Answers are in a **separate file**.

LESSON	1-1	1-2	1-3
Questions touching the lesson	8	19	23

طريقة الاستخدام

الأسئلة على نمط الكويز والميدتيرم: **تصنيف مع تبرير**، **بناء جدول أو رسم**، **قراءة رسم واستخراج قيمة**، و**اكتشاف خطأ طالب**. حلّها مغلق الكتاب ثم صحّح من ملف الإجابات. الجزء F محاكاة لجزء هذا الفصل في الكويز الصفي الأول.

Part A · Concept and interpretation (multiple choice)

Give the letter and a one-line reason, as in a written exam.

A1 1-1 CLO1-A

A researcher asked 400 Muscat households about car ownership and reports: "62% of the households surveyed own two or more cars." This statement is:

(a) inferential, because it comes from a sample

(b) descriptive, because it only describes the 400 households surveyed

(c) inferential, because it is a percentage

(d) neither, because no graph is used

A2 1-1 CLO1-A

From the same 400 households the researcher writes: "About 62% of **all** Muscat households own two or more cars." This statement is:

(a) descriptive

(b) inferential

(c) a census result

(d) a parameter computed from the population

A3 1-1 CLO1-A

The SQU registrar computes the mean GPA of **every** 2025 graduate. Which description is correct?

(a) a survey producing a statistic

(b) a census producing a parameter, summarised with descriptive statistics

(c) inferential statistics, because a mean was calculated

(d) a biased sample

A4 1-2 CLO1-B

Which variable is **qualitative**?

(a) number of absences in a semester

(b) student ID number

(c) waiting time at a clinic

(d) annual rainfall in Salalah

A5 1-2 CLO1-B

The **year** a car was manufactured is measured at the:

(a) nominal level

(b) ordinal level

(c) interval level

(d) ratio level

A6 1-2 CLO1-B

Which statement is meaningful **only** because the variable is at the ratio level?

(a) 30 °C is twice as hot as 15 °C

(b) a 6-hour flight lasts twice as long as a 3-hour flight

(c) an IQ of 140 is twice as intelligent as 70

(d) grade A is twice as good as grade C

A7 1-2 CLO1-B

Customers rate a service 1 = very poor, 2 = poor, 3 = fair, 4 = good, 5 = excellent. The level of measurement is:

(a) nominal

(b) ordinal

(c) interval

(d) ratio

A8 1-2 CLO1-B

The number of patients visiting a clinic each day is:

(a) qualitative, nominal

(b) quantitative, continuous, interval

(c) quantitative, discrete, ratio

(d) quantitative, discrete, interval

A9 1-3 CLO1-C

A traffic officer checks every 20th car leaving a mall car park, starting with a randomly chosen car among the first 20. The method is:

(a) simple random
(b) systematic
(c) stratified
(d) cluster

A10 1-3 CLO1-C

SQU randomly selects 3 of its colleges and surveys **every** student in those colleges. The method is:

(a) stratified, because colleges are groups
(b) cluster
(c) systematic
(d) convenience

A11 1-3 CLO1-C

A TV programme asks viewers to send an SMS to vote on a new law. The sample is:

(a) simple random, because anyone can vote
(b) a volunteer (self-selected) sample, likely biased
(c) cluster
(d) a census of viewers

A12 1-3 CLO1-C

In a random sample of 200 students, 48% support a policy; the true figure for all students is 51%. The 3% difference is best described as:

(a) nonsampling error
(b) sampling error
(c) a biased sample
(d) a recording error

A13 1-3 CLO1-C

A clinic's scale is miscalibrated and adds 0.5 kg to every weight. This produces:

(a) sampling error
(b) nonsampling error
(c) no error, because it is the same for everyone
(d) a cluster effect

A14 1-3 CLO1-C

Which feature is **essential** to a stratified sample?

(a) equal numbers from each stratum
(b) a random sample drawn from every stratum
(c) only some strata are sampled
(d) every member of a stratum is included

Part B · Short exam problems by lesson

Classification with justification, Table D, systematic k, find-the-mistake.

B1 1-1 CLO1-A

To study daily study hours of all 3,500 first-year SQU students, a researcher asks 150 randomly chosen first-year students; their mean is 2.4 hours. Identify (a) the population and N, (b) the sample and n, (c) the variable with its type and level, (d) whether 2.4 is a parameter or a statistic, (e) the branch of statistics if the researcher concludes "first-year students study about 2.4 hours a day".

B2 1-1 CLO1-A

Census or survey? Justify each. (a) The NCSI collects data on every person, household and building in Oman. (b) SQU emails a questionnaire to 300 randomly chosen students. (c) A teacher records the marks of every student in her class in order to describe that class.

B3 1-2 CLO1-B

Give the type (qualitative or quantitative, discrete or continuous) and the level of measurement of each: (a) governorate of residence (b) number of siblings (c) body temperature in °C (d) customer rating poor / fair / good / excellent (e) time to finish an exam, in minutes (f) Oman Air flight number WY101.

B4 1-2 CLO1-B

Find the boundaries of each recorded value: (a) 48 kg (b) GPA 3.7 (c) 0.125 s.

B5 1-2 CLO1-B · CLO3-C

"Muscat's temperature rose from 20 °C in January to 40 °C in June, so June is twice as hot." Evaluate the claim.

B6 1-2 CLO1-B

Two students describe "number of goals scored in a match". Student A: "quantitative, continuous, because it can be large." Student B: "quantitative, discrete, ratio." Who is right, and what is wrong with the other answer?

B7 1-3 CLO1-C

A company has N = 750 employees numbered 001–750. Select a simple random sample of n = 6 using Table D, starting at row 2, column 1 (06512 07836 88456 36313 30879 51323 76451 25578 15986 50845 57015 53684 57054 93261).

B8 1-3 CLO1-C

A list of 250 patients is to be sampled systematically, n = 12. (a) Find k and list the sample if the random start is 7. (b) A student rounds k up to 21 and the random start happens to be 21. What goes wrong?

B9 1-3 CLO1-C · CLO3-C

A researcher wants to compare study habits of male and female SQU students, but females are about two-thirds of the students. Explain why a stratified sample is better than a simple random sample here, and describe how to take it.

B10 1-3 CLO1-C

Name the sampling method and say whether it is likely to be biased: (a) the first 30 shoppers to enter Lulu on Saturday morning (b) an online poll on a news website that anyone can answer (c) the names of all 2,000 employees are placed in a box and 40 are drawn.

B11 1-3 CLO1-C

The Ministry of Education wants grade-10 maths scores. Plan P: randomly pick 10 of the 120 schools and test every grade-10 student in them. Plan Q: in every one of the 120 schools, randomly pick 5 grade-10 students. Name each plan and give one advantage of each.

B12 1-3 CLO1-C

Find the mistake. For 86 trees and n = 6, a student used k = 15 and a random start of 13: "13, 28, 43, 58, 73, 88".

B13 1-3 CLO1-C

Find the mistake. For N = 4998 laptops, a student read Table D row 2 in 5-digit groups and chose laptops 06512 and 07836 first.

B14 1-1 1-3 CLO1-A · CLO1-C · CLO3-C

A website surveys 50 members of a "sleep well" club, who volunteered, and concludes: "Students who sleep more get higher GPAs." (a) Which branch of statistics is the conclusion? (b) Which sampling method was used? (c) Give two reasons to doubt the conclusion.

Part C · Long structured questions

Midterm-style, multi-part. Marks in brackets.

C1 1-2 CLO1-B **[9 marks]**

An SQU hospital record contains: (a) name (b) age in years (c) weight in kg (d) condition: critical / serious / fair / good (e) date of admission (f) number of previous visits (g) ward number (h) body temperature in °C (i) blood type. For each, state qualitative or quantitative (and discrete or continuous) and the level of measurement.

C2 1-1 1-2 1-3 CLO1-A · B · C **[10 marks]**

The Oman Traffic Authority wants to know the average daily driving time of licensed drivers in Oman. It plans to ask a sample of 1,200 drivers.

1. State the population, the sample and the variable, with the variable's type and level.
2. Describe a stratified design and a cluster design for this study.
3. Give one possible nonsampling error.
4. Classify: (i) "The 1,200 drivers averaged 74 minutes a day." (ii) "Omani drivers spend between 70 and 78 minutes a day driving."

C3 1-3 CLO1-C **[10 marks]**

A college lists its 480 students, numbered 001–480. Table D row 2: 06512 07836 88456 36313 30879 51323 76451 25578 15986 50845 57015 53684 57054 93261.

1. Select a simple random sample of 8 students starting at row 2, column 1.
2. Select a systematic sample of 8: find k, and take the random start as the first 2-digit number in row 2 that is between 01 and k.
3. The college wants both male and female students represented. Which method should it use, and how?

C4 1-2 CLO1-B · CLO3-C **[8 marks]**

A survey records: (i) gender, (ii) satisfaction coded 1–5, (iii) room temperature in °C, (iv) monthly income in OMR. For each variable, state the level and say whether each comparison is meaningful: "is different from", "is greater than", "is 10 units more than", "is twice".

Part D · True / false with justification

The justification is the mark.

D1 1-2 CLO1-B · CLO1-C

True or false? Justify. Every quantitative variable is measured at either the interval or the ratio level.

D2 1-2 CLO1-B · CLO1-C

True or false? Justify. A discrete variable cannot be at the ratio level.

D3 1-3 CLO1-B · CLO1-C

True or false? Justify. A census has no sampling error.

D4 1-3 CLO1-B · CLO1-C

True or false? Justify. Taking a larger sample removes nonsampling error.

D5 1-3 CLO1-B · CLO1-C

True or false? Justify. In cluster sampling, every cluster contributes some members to the sample.

D6 1-3 CLO1-B · CLO1-C

True or false? Justify. In stratified sampling, the same number must be taken from each stratum.

D7 1-2 CLO1-B · CLO1-C

True or false? Justify. Postal codes are quantitative because they are numbers.

D8 1-3 CLO1-B · CLO1-C

True or false? Justify. A convenience sample becomes unbiased if it is large enough.

Part E · Dr. Ahmed's review self-study exercises

The two exercises that close his deck.

E1 1-2 CLO1-B

Review exercise (Dr. Ahmed). Name the level of measurement: (a) ranking of data, a precise difference between units of measure, and a true zero; (b) mutually exclusive categories that can be ranked, but without precise differences between ranks; (c) mutually exclusive categories with no order or ranking; (d) ranking and precise differences, but no true zero.

E2 1-2 CLO1-B

Review exercise (Dr. Ahmed). An SQU hospital patient record shows: (a) Name: Ibrahim Ahmed (b) Age: 35 years (c) Condition: Fair (d) Date of admission: May 15 2019. State the type of data and the level of measurement of each.

Part F · In-Class Quiz I simulation, Chapter 1 part (15 marks · 15 minutes)

Closed book.

F1 1-1 CLO1-A

[2 marks] Define *inferential statistics* and give an example from Oman.

F2 1-2 CLO1-B

[4 marks] Type and level: (a) number of credit hours registered (b) Likert response "agree" (c) graduation year (d) distance from home to SQU in km.

F3 1-3 CLO1-C

[4 marks] Name the method: (a) every 15th invoice is audited (b) 4 of 11 governorates are chosen at random and all hospitals in them are surveyed (c) 10 nurses are randomly chosen from each department (d) a lottery draws 25 student IDs.

F4 1-3 CLO1-C

[3 marks] N = 92 and n = 7 for a systematic sample. Find k and the sample if the random start is 9.

F5 1-2 CLO1-B

[2 marks] Find the boundaries of 7.25 g and of 120 cm.

Chapter 1 study set: 7 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's slide pack, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief · 01 · Lesson 1-1: Descriptive and Inferential Statistics · 02 · Lesson 1-2: Variables and Types of Data · 03 · Lesson 1-3: Data Collection and Sampling · 04 · Revision pack · **05 · Exam-style questions** · 06 · Full worked answers